

Peer Review

Review of: "ToxiLab: How Well Do Open-Source LLMs Generate Synthetic Toxicity Data?"

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1. Independent researcher

1. What the work claims

The paper asks whether open-weight LLMs can stand in for GPT-4 as generators of synthetic toxic text for training moderation classifiers. It proceeds in two stages. In Stage 1 (§3.1, §5.1, Table 2), seven checkpoints from six families (LLaMa-7B, LLaMa-13B, Vicuna-13B, Mistral-7B, Falcon-7B, Bloom-7B, Gemma-7B) are prompted with a Role/Requirement/Few-shot template, including an explicit jailbreak persona ("AOA (Absolutely Obedient Agent)", Appendix A, V1 – Positive), and scored on a "success rate" of valid positive samples for the Political and Hate categories plus a binary human-quality check. In Stage 2 (§3.2, §5.2), Mistral is LoRA-fine-tuned with LLaMA-Factory on 3,000 positive and 3,000 negative examples per configuration, either on one category (Table 4) or on a blend of hate, violence, and sex data, optionally with GPT-generated positives ("Mixture" and "Mix_GPT", Table 3). The generated data are appended to the training set and judged by the F1 and accuracy of a two-layer MLP classifier on a 2,500-item evaluation split per category (§4.1, §4.3).

The headline conclusions (Abstract; §5.3–5.5; §7) are that Mistral is the best open generator, that fine-tuning "significantly enhances data reliability and diversity" and reduces hallucination, that mixing categories improves generalization, that performance plateaus after epoch 3, and that the fine-tuned models have been deployed by "a leading cloud provider" (§6). The practical question is a good one. Industrial teams do face exactly this choice, and the authors have access to multi-category, multi-round annotated data that few academic groups have. The prompt texts in Appendix A are given verbatim, which I appreciate; many papers in this area hide them.

2. Whether the conclusion holds on the data

2.1 The missing control. I could not find, in any table, a classifier trained on the real training data alone. The algorithm (§3.3) defines $D_{\text{augmented}}$ as D_{train} plus D_{harmful} , and every row of Tables 3, 4, 5, and 7 appears to be some augmented variant, or a prompt-only "Mistral" row, or a "GPT4(baseline)" row. Without the no-augmentation row, it is impossible to say whether any synthetic data helped at all, let alone whether fine-tuned generation helps more than prompting. This is the single comparison the whole paper rests on. The number of synthetic samples added per row is also not stated, so rows may differ in training-set size as well as in data source.

2.2 The text contradicts its own tables on GPT-4. §5.3 says: "Although GPT-4 remained the strongest performer, fine-tuned Mistral approached its performance." Table 3 says otherwise. On Hate, F1: Mixture (Epoch 5) 0.672 against GPT4(baseline) 0.628. On Sex, F1: Mixture (Epoch 3) 0.835 against 0.628. On Violence, F1: Mixture (Epoch 5) 0.713 against 0.649. On Self-harm, F1: Mix_GPT (Epoch 5) 0.788 against 0.583. In Table 4, block Sex, Sex_epoch5 reaches F1 0.755 against GPT4(baseline) 0.591. If these numbers are right, the fine-tuned open model beats GPT-4 in most cells, and the paper undersells its result. If the sentence is right, the tables are wrong. One of them has to change.

2.3 The same GPT-4 cell has two values. Table 3, row GPT4(baseline), columns Sex: F1 0.628, Acc 0.721. These are identical to the Hate columns of the same row (0.628, 0.721), which already looks like a copy-paste. Table 4, block Sex, row GPT4(baseline) gives F1 0.591, Acc 0.685 for what should be the same experiment. The Mistral rows, by contrast, agree between the two tables (Hate 0.608/0.706, Sex 0.534/0.676), so I think Table 3's Sex entry for GPT-4 is wrong. Every Sex comparison against GPT-4 in §5.3 depends on which value is correct.

2.4 Single-category fine-tuning on Hate makes things worse, and the text says the opposite. Table 4, block Hate, F1: Mistral (prompt-only) 0.608; hate_epoch1 0.540; hate_epoch2 0.587; hate_epoch3 0.563; hate_epoch5 0.590. Every fine-tuned row is below the prompt-only row, on accuracy as well (0.706 against 0.661–0.678). Yet §5.3 cites Table 4 for "improved F1 scores and accuracy across multiple datasets", and §5.5 cites it for "Fine-tuned models consistently outperformed those relying solely on prompt-generated data". Table 4 contains two datasets. On one of them, the claim holds (Sex); on the other, it fails.

2.5 The epoch story is not in Table 4. §5.5 states that improvements "plateaued beyond epoch 3, and some models experienced performance degradation at epoch 5 (Table 4)", and §5.4 attributes "data duplication and overfitting, particularly in later training epochs" to the same table. In Table 4, epoch 5 is

the best epoch in both blocks (Hate F1 0.590, Sex F1 0.755), and Sex rises monotonically from 0.587 to 0.755. Table 4 has no duplication column at all. The only degradation with epochs I can find is in Table 3, Self-harm, F1: Mixture 0.593 (Epoch 1), 0.577 (Epoch 3), 0.541 (Epoch 5), which is a category not included in the Mixture training blend (§5.2). That is an interesting out-of-category signal, but it is not what the text describes.

2.6 Diversity, hallucination, and reliability are asserted, not measured for the fine-tuned models. The abstract and §5.4 claim that fine-tuning improves diversity, reduces hallucination rates, and produces "more subtle and realistic toxic content". The only diversity-related measurement in the paper is Table 6 (duplication rates), and its title and Appendix C heading place it in the prompt-engineering stage. I found no hallucination metric, no definition of hallucination for this task, no lexical or embedding diversity statistic, and no human rating of subtlety for any fine-tuned output. The five generation-error cases and three classification-error cases announced in §5.6 are not present in the Qeios rendering; as rendered on Qeios, those examples are not legible, and Figure 2 (ablation on data mixing) is an image I could not read numerically.

2.7 Stage 1 cannot support model selection as reported. Table 2 gives bounds (" $\geq 85\%$ ", " $\leq 10\%$ ") rather than rates, with no sample size. §5.1 says a "minimum threshold of 60% of the total samples was required", yet Gemma-7B is listed at $\geq 50\%$ (Political) and still appears as an evaluated model, while LLaMa-7B, LLaMa-13B, and Bloom-7B are $\leq 10\%$ on Hate. The Human Eval columns are check marks. The text says a check means annotators reached "a high Fleiss' kappa score", but neither the kappa, nor the number of annotators, nor the number of items, nor the rating scale is given. §5.4 names Bloom among models with "frequent refusals"; Table 2 records a refusal only for Gemma-7B ("Reject to Answer"). GPT-4 has no row in Table 2, so RQ1's "compared to proprietary models" is not answered at the generation stage. Finally, §4.2 says Mistral and Vicuna were both fine-tuned, but no Vicuna fine-tuning result appears anywhere.

2.8 Several tables contain arithmetically impossible cells. I recomputed F1 as $2PR/(P+R)$ for every row that reports precision and recall.

- Table 7, block Sex, row sex_v3: P 0.705, R 0.668, F1 0.653. The harmonic mean is $2 \times 0.705 \times 0.668 / 1.373 = 0.686$. A reported F1 below both P and R cannot occur.
- Table 7, block Sex, row hate_sex_v3: P 0.628, R 0.664, F1 0.664; recomputed 0.646.
- Table 7, block Political Figure, row multi_hate_v1: P 0.938, R 0.962, F1 0.971; recomputed 0.950. F1 above both P and R is impossible.

The other twelve rows of Table 7 reproduce to three decimals, so these three are specific errors, not a different averaging convention. Table 5 (Appendix A) has a column-header problem instead: row 1 reads Latency(s) 0.174, Precision 0.440, Recall 0.250, F1 0.712. Those cannot be P, R, and F1 ($2 \times 0.440 \times 0.250 / 0.690 = 0.319$). Read as P = 0.174, R = 0.440, F1 = 0.250, and a final accuracy column, the row is consistent ($2 \times 0.174 \times 0.440 / 0.614 = 0.249$), and the same shift works for rows 3–5. Row 3 then matches Table 7, block Hate, row Mistral_V1 exactly (0.305, 0.151, 0.202, 0.870). So the "Latency(s)" header is misplaced, and no latency is actually reported, which matters because the paper's cost-effectiveness claim has no other numeric support.

2.9 The ablation conclusions in Appendix D.1 do not follow from Table 7. The text says mixed data "generally improves the recall and F1 scores across most datasets" and that single-category models "show high precision but lower recall compared to those fine-tuned on mixed data." In block Sex, multi_hate_v1 has lower recall (0.668 against 0.720) and lower F1 (0.626 against 0.663) than sex_v1, and also lower F1 than the Mistral_V3 reference (0.634). In block Hate, hate_v1 has lower precision than multi_hate_v1 (0.175 against 0.241), the reverse of the stated pattern. The mixed model's accuracy on Hate (0.835) is below the prompt-only reference (0.870). The reference rows themselves differ between blocks (Mistral_V1 for Hate, Mistral_V3 for Sex and Political Figure), so the comparator is not held fixed. The prompt-version conclusion is circular: Appendix D opens with "the effects of different prompt versions are considered the same across all experiments" and then concludes that prompt version "has a minimal impact".

2.10 Two evaluation regimes are mixed without explanation. Hate F1 is around 0.54–0.67 with accuracy around 0.66–0.74 in Tables 3 and 4, but around 0.17–0.25 with accuracy 0.81–0.87 in Tables 5 and 7. High accuracy with F1 near 0.2 suggests a strongly imbalanced test set in the appendix experiments, while §4.1 describes one 2,500-entry evaluation set per dataset. The reader needs to know which test set, with what positive rate, underlies each table.

2.11 No uncertainty is reported. §4.3 says the MLP was trained "with different random seeds," but no table gives a mean, a standard deviation, or the number of seeds. A rough scale: for a single run at accuracy near 0.7 on $n = 2,500$, the binomial 95% half-width is $1.96 \times \sqrt{(0.7 \times 0.3 / 2500)} = 1.96 \times 0.00917 \approx 0.018$. The Hate accuracy gap between Mistral (0.706) and GPT4(baseline) (0.721) in Table 3 is 0.015, inside that band, and so are several of the epoch-to-epoch differences in Table 4 (for example, Sex, Acc, epoch2 0.731 against epoch3 0.740). The paired design would tighten this somewhat, but without seed variance, I cannot tell which differences are real.

2.12 Deployment and cost claims have no data. §6 reports that the models were deployed and achieve "strong performance in diverse online environments," with no metric, time window, traffic volume, baseline, or even the task the deployed model performs. The section also slides from generators to detectors ("fine-tuned open-source LLMs can effectively identify and flag harmful content"), and §7 does the same ("Fine-tuned models struggled with detecting nuanced harmful language"), although the fine-tuned models in this paper generate data and the detector is the MLP. "Cost-effective" (Abstract, §1) is not accompanied by any cost or latency comparison with GPT-4. As written, §6 should either carry numbers or be reduced to a sentence.

Taken together, the defensible conclusions from the rendered data are narrower than the abstract: on Sex, Violence, and Self-harm, data from a Mistral model fine-tuned on a category blend give a higher MLP F1 than data from prompt-only Mistral, and apparently than GPT-4 data, in single runs; on Hate, the picture is mixed, and single-category fine-tuning did not help. That is still a useful finding. It needs the missing control and error bars before it can carry the stronger claims.

3. Reproducibility

Some things are well documented. Appendix B gives the framework (LLaMA-Factory), hardware (4 A100 GPUs on Azure), batch size 16, gradient accumulation 8, cutoff length 4,096, learning rate $1e-5$, warmup ratio 0.1, AdamW with weight decay 0.01, dropout 0.1, and fp16. Appendix A gives the three prompt versions in full. The MLP architecture is stated (two hidden layers of 600 units, ReLU, Adam). The rest is missing, and for an LLM paper, the gaps are the usual ones.

Data. All five datasets are proprietary (§4.1), sizes are approximate ("~10k", Table 1), and neither the real data, the synthetic data, nor the 2,500-item evaluation splits are released. I understand the constraints on toxic and customer-derived data. Even so, a public benchmark run (the paper already cites ToxiGen) would let others check the method.

Model identities. No checkpoint names or versions are given for any open model (base or instruct; Mistral-7B v0.1 or later; LLaMa vs LLaMA-2, where the text says LLaMA2 but reference [26] is the original LLaMA paper). §4.2 switches from Mistral to "Mixtral", Appendix C is titled "Prompt engineering mixtral", and §5.6 mentions "Mixtral_Mixture3"; Mistral-7B and Mixtral are different architectures, and I could not determine which was fine-tuned. For GPT-4, no snapshot, access date, or API endpoint is given.

Generation settings. Temperature, top-p, repetition penalty, number of samples per prompt, and total number of generated samples are not reported for either stage. The only generation-related setting is

maximum new tokens (4,096, Appendix B), while the prompts ask for 30 items per call. The meaning of "hate_30_30" and of Sample = long/short (Table 5, Table 6) is never defined.

LoRA and training. Rank, alpha, target modules, and number of training steps are absent. Whether batch size 16 is per device or global is unclear.

Downstream classifier. The MLP's input representation is not stated anywhere: bag of words, TF-IDF, or sentence embeddings from some encoder. This alone makes Tables 3, 4, 5, and 7 unreproducible. Number of seeds, training epochs, and the decision threshold are also missing.

Code, seeds, and deduplication. No code repository is linked. No seeds are given. §3.3 says duplicates in D_harmful are removed but not by what criterion (exact string, normalized string, near-duplicate threshold). Table 6 is hard to interpret without the denominators: taking Table 5's Num column as denominators for the hate rows, most rates reproduce ($355/3034 = 11.70\%$, $137/3106 = 4.41\%$), but row 1 normal gives $21/1520 = 1.38\%$ against the reported 1.30%, row 2 gives $390/2917 = 13.37\%$ (reported 13.30%) and $58/3033 = 1.91\%$ (reported 1.80%), and row 5 normal gives $110/3000 = 3.67\%$ (reported 3.60%). I am assuming those denominators; the paper should state them. In the same table, sex_30_30 V1 long, normal, has Dup Mean for Fixed ID 9.54 with Dup Num for Fixed ID 105, while every other row satisfies Dup Num = Dup Mean $\times$ 50 or $\times$ 100; and sex_30_30 V3 short, harmful, reports the mean as "0%", a unit the column uses nowhere else.

4. What specifically to fix

1. **Tables 3 and 4: add a no-augmentation row** (MLP trained on D_train only) for every category, and state how many synthetic samples each row adds. Without it, the paper cannot claim that synthetic data helps.
2. §5.3, sentence "Although GPT-4 remained the strongest performer": reconcile it with Table 3 (Mixture Epoch 5 Hate F1 0.672, Mixture Epoch 3 Sex F1 0.835, Mixture Epoch 5 Violence F1 0.713, Mix_GPT Epoch 5 Self-harm F1 0.788, each above GPT4(baseline)) and Table 4 (Sex_epoch5 F1 0.755 against 0.591).
3. Table 3, row GPT4(baseline), Sex columns (0.628/0.721) duplicate the Hate columns and disagree with Table 4 (0.591/0.685). Correct whichever is wrong.
4. §5.3 and §5.5: withdraw "consistently outperformed" for Table 4, where all four hate_epoch rows (F1 0.540–0.590) fall below prompt-only Mistral (0.608). Report it as a negative result; it is informative.

5. §5.4 and §5.5: either remove the epoch-3 plateau and epoch-5 degradation claims or point them to the evidence that shows them. Table 4 shows epoch 5 as best in both blocks. If the claim refers to Table 3 Self-harm, say so and note that Self-harm was outside the Mixture blend.
6. Abstract and §5.4: the words "hallucination," "reliability," and "diversity" need a metric each, computed on fine-tuned outputs (for instance distinct-n, self-BLEU, or embedding dispersion, and a human-rated hallucination or off-target rate with inter-annotator agreement). Otherwise, drop them from the abstract.
7. Table 7: fix the three impossible F1 values (Sex/sex_v3 0.653, recomputed 0.686; Sex/hate_sex_v3 0.664, recomputed 0.646; Political Figure/multi_hate_v1 0.971, recomputed 0.950). Use one reference row across blocks, and align the name "hate_multi" in the text with "multi_hate_v1" in the table.
8. Appendix D.1, the four "trends": rewrite after the corrections. The Sex block and the Hate precision values currently contradict the first and third bullets. Remove the prompt-version conclusion, or run the prompt versions separately instead of assuming they are equivalent.
9. Table 5 header: the "Latency(s)" label is shifted; the columns appear to be Precision, Recall, F1, Accuracy. Also, disambiguate the two rows both labelled "hate_30_30 / V1 / long".
10. Table 2: give actual success rates with denominators, not bounds. Report Fleiss' kappa, annotator count, and item count behind each check mark. Explain why Gemma-7B at $\geq 50\%$ was evaluated despite the stated 60% threshold, and add a GPT-4 row.
11. §4.2: say which model was fine-tuned (Mistral-7B or Mixtral), give exact checkpoint identifiers, and either report the Vicuna fine-tuning results or remove Vicuna from the Stage 2 description.
12. §4.1 and Tables 5/7: state the positive rate of each evaluation set and explain why Hate accuracy is 0.81–0.87 at F1 around 0.2 in the appendix but about 0.7 at F1 around 0.6 in Tables 3–4.
13. All result tables: report mean $\pm$ standard deviation over the seeds mentioned in §4.3, with the number of seeds, and a paired significance test (McNemar or bootstrap on the 2,500 items) for the main comparisons.
14. Appendix B and §4.3: add LoRA rank, alpha, and target modules; decoding parameters for generation; total generated samples; the MLP input features; and the deduplication criterion.
15. §6: provide deployment metrics or shorten it to a statement of fact. Also, fix the generator/detector confusion here and in §7.
16. §5.6: the promised generation-error and classification-error examples do not appear in the Qeios version. Restore them, with appropriate masking.

17. Table 6: give denominators, explain "Fixed ID," fix the sex_30_30 V1 long normal row (9.54 against 105), and the "0%" entry.
18. §1, first contribution ("one of the first (if not the first)"): qualify it against the generation-based augmentation work the paper itself cites in [1], [4], and [6], stating exactly what is new here.
19. §9 and the separate "Ethics Statement": merge them. Say whether the generated data or LoRA weights will be shared under access control, how the "company service collections" in §4.1 were consented to or anonymized, and what support was given to annotators exposed to this material. The Political category ("content that is targeting public figures") deserves a specific word on misuse.

5. Recommendation

Major revisions. The question matters for practitioners, the data are unusually rich, and the Sex, Violence, and Self-harm results in Table 3 may well show that a fine-tuned open model can replace GPT-4 as a generator. I would like to see that result stand. At present, though, the paper lacks the no-augmentation control, reports no variance, contains internally contradictory GPT-4 numbers and three impossible F1 values, and states several conclusions (GPT-4 strongest, consistent gains, epoch-3 plateau, reduced hallucination, deployment success) that its own tables either do not support or contradict. None of this needs new ideas. It needs the control run, seed statistics, corrected tables, and a text that says what the numbers say.

Declarations

Potential competing interests: No potential competing interests to declare.